CO6602 - Cryptography and Security

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Check-in code:

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Assessment Dates

Component	Weighting %	Learning outcome(s) assessed	Assessment category
Assessment 1: In-class Test	40%	1, 2, 4	In-class Test
Assessment 2: Individual Report (2,400 words)	60%	3, 4, 5, and 6	Coursework

Assignment 1 Due date: **30th October 2026** – 13: 00 Hrs;
Assignment 2 Due date: **12th December 2026** - 13:00 Hrs;

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Week 1 – Lesson

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Objectives

1. Briefly outline the history, evolution and impart of Cryptography
2. Describe the key security requirements of confidentiality, integrity, and availability.
3. Discuss the types of security threats and attacks that must be dealt with and give examples of the types of threats and attacks that apply to different categories of computer and network assets.
4. Provide an overview of keyless, single-key, and two-key cryptographic algorithms.
5. Outline at least three physical security controls

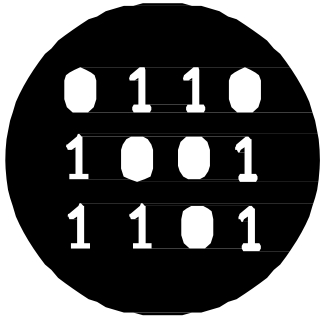
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History of Cryptography: Why Secrets Matter

- Cryptography protects information when communication channels cannot be trusted.
  - It keeps a message readable only to the intended audience.
  - It lets people share sensitive information with confidence.
- Its history is an ongoing contest between codemakers and codebreakers.
  - **Codemakers** strengthen methods as **codebreakers** find weaknesses.
  - Each breakthrough shifts the advantage, then sparks new defenses.
  - Changes in language, mathematics, machines and computing repeatedly alter the balance
- It influenced wars and state decisions by protecting intelligence.
- It also underpins trade and personal privacy in daily life.

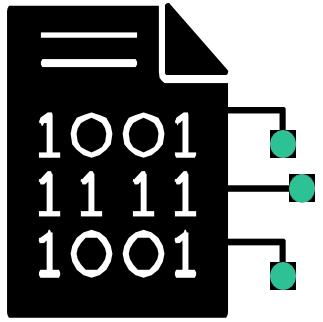
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History of Cryptography: Hiding Message



Steganography: hide the signal

- It conceals that a message exists, so observers see nothing suspicious.
- Classic tricks include hiding text inside objects or everyday actions.



Cryptography: hide the meaning

- It keeps the message visible, but makes the content unreadable.
- Only someone with the right key can restore the original meaning.

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The Story of Cryptography Since Ancient Times



Alice

Let us meet in the alley today



Eve

Let us meet in the alley today

key



Bob

XYZGKI UKNJK SSJVF GU

key

E

C

C'

D

Let us meet in the alley today

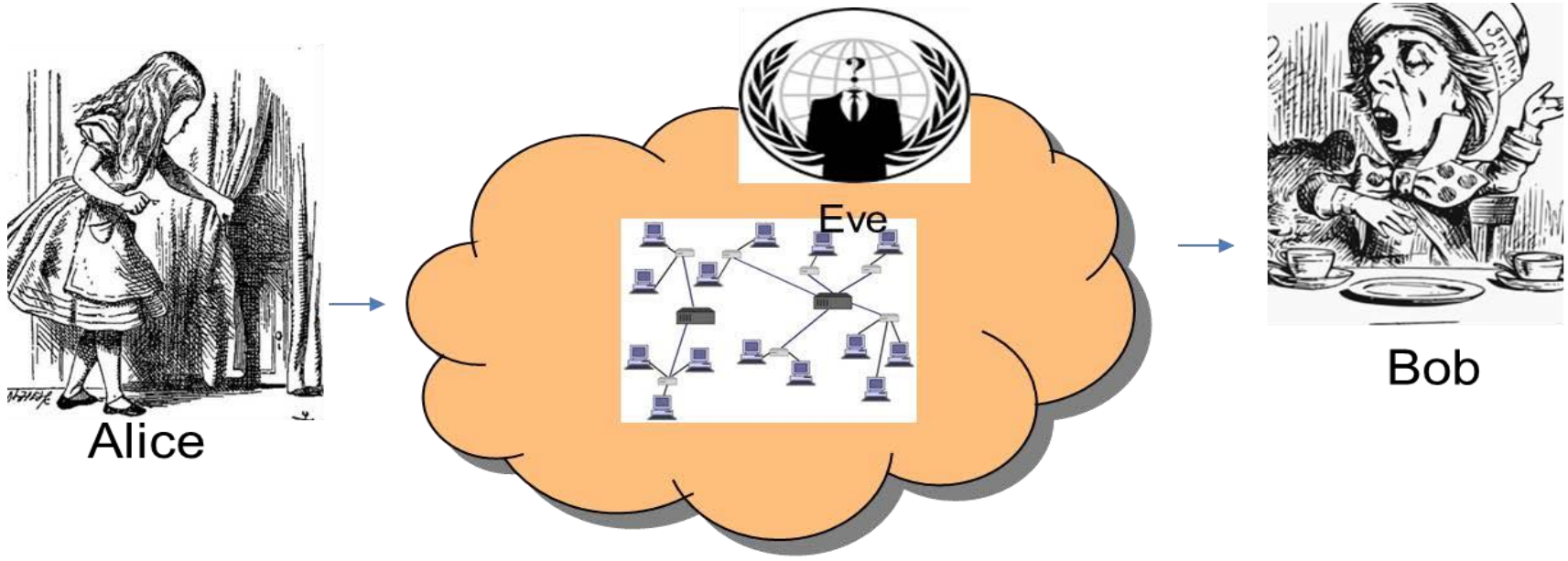
How do they agree on the "key"? -Chicken and Egg Problem

Notations:

- C – Sent ciphertext
- C' – Received ciphertext
- D – Decryption
- E – Encryption
- C – Ciphertext

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Cryptography in Modern Times



Alice and Bob cannot meet in advance for every situation

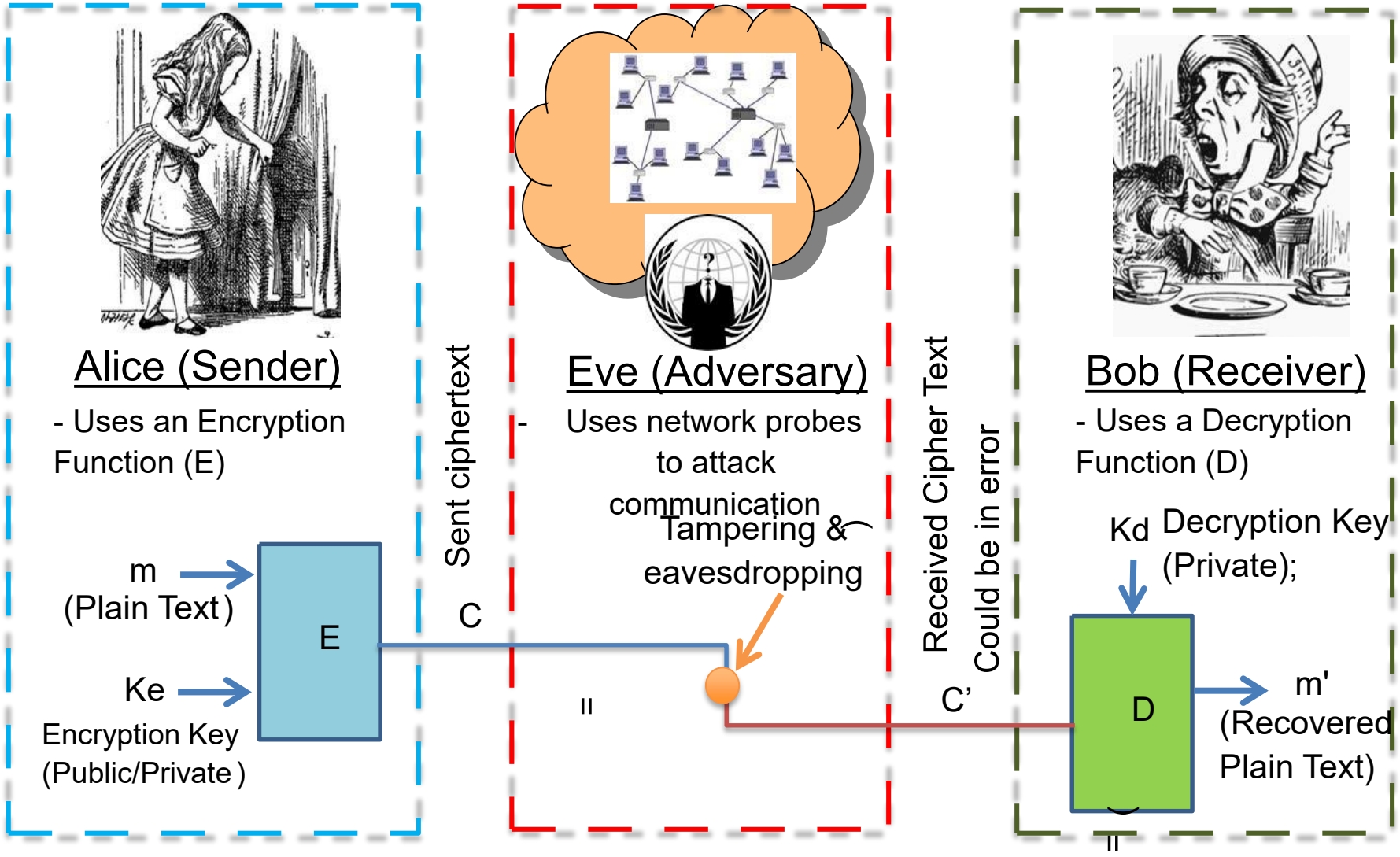
Can Mathematics come to their aid?

The magic tool is the One Way function.
We will consider many such functions based on numbers in the subject

Images: General Internet resources

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Alice and Bob terms with Notations



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History of Cryptography: Evolution of Ciphers

Ancient

Hidden messages, scytales and substitutions

9th c.

Al-Kindi and frequency analysis

16th c.

Vigenère and polyalphabetic ciphers

1918

One-time pad and Enigma

1970s

Public-key cryptography and RSA

Future

Quantum computing and quantum keys

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History of Cryptography: Early Ciphers

- Transposition: shuffle the order
  - Letters stay the same, but their positions are rearranged.
  - It hides readable words by scrambling their sequence.
  - Examples include rail-fence methods and the Spartan scytale.
  - Language frequencies remain present but are rearranged.
- Substitution: swap the symbols
  - Each letter is replaced with a different letter or sign.
  - It protects meaning by changing what each character stands for
  - Examples include Caesar shift and arbitrary alphabets.
    - A simple, systematic method that shaped early encryption thinking.
    - A fixed shift moves every letter the same number of steps.
  - Security depends on the mapping and how it is used.

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Historical Impact and Evolution

Mary, Queen of Scots and the Babington Plot – Story extracted from the Code Book

- A cipher in a power struggle:
  - Mary, Queen of Scots relied on secret letters to coordinate a plot.
  - Mary's correspondence used a nomenclator with alphabet symbols, codewords, nulls and doubling marks.
- Interception turns secrecy into evidence
  - The messages were intercepted and deciphered by Elisabeth I's codebreakers.
- A decoded message changes the outcome
  - The decrypted text was used against her, leading to execution



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History: Babbage and the break of Vigenère

- Charles Babbage identified how repeated ciphertext segments can reveal key length.
- The method was later published independently by Friedrich Kasiski.
- Once the period is known, the text separates into several monoalphabetic ciphers.



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History of Cryptography: Vigenère Broken

- Frequency analysis wins (9th c.)
- Single-alphabet ciphers become vulnerable

- Vigenere introduces multiple alphabets
- Shifting alphabets blur letter-frequency patterns

- The 'unbreakable cipher' reputation
- Believed secure for centuries in practice

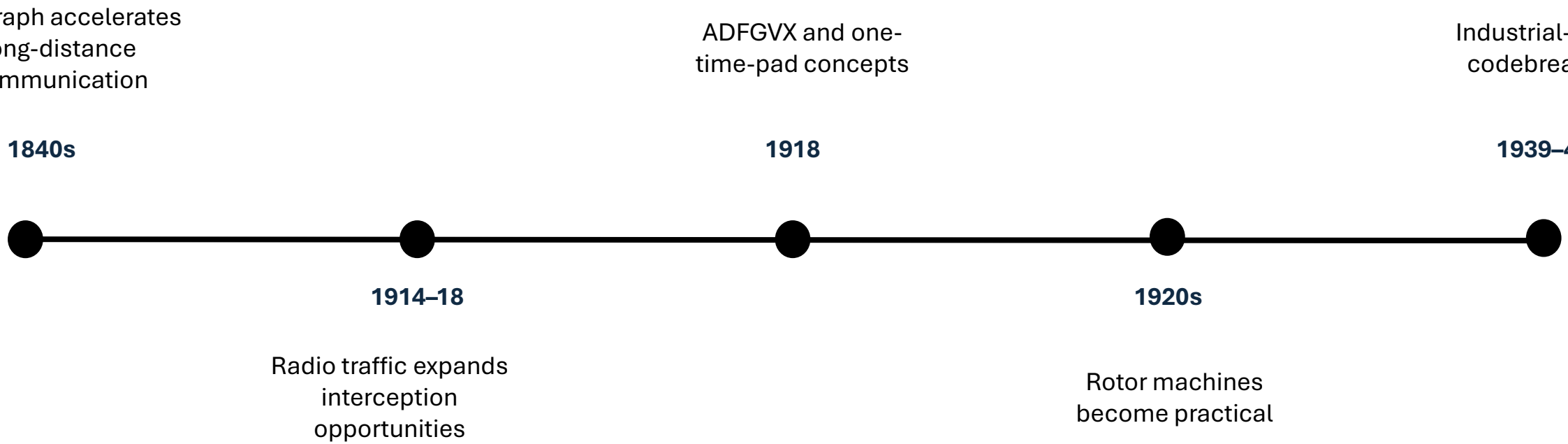
- Babbage finds a structural weakness (1800s)
- Cracking becomes possible with the right method

- Advantage swings back to codebreakers
- Confidence in permanence of security is shaken



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History: From Telegraph to Machine Cryptography



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Cryptography is the art and science of keeping information secret. As communication technology evolved, so did the methods to protect it.

The History of Cryptography

From Telegraph to Machine Cryptography

SAME GOAL. A SAFER TOMORROW.

"From dots and dashes to digital defenses — cryptography has always protected what matters."

1. Early Ideas (Ancient Times – 1800s)

The roots of secret communication.

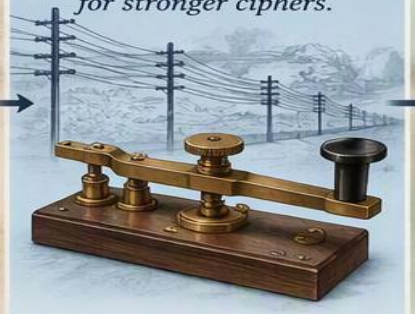


- Egyptions used non-standard hieroglyphs.
- Greeks used the Scytale (a wrapped strip of parchment).
- Julius Caesar used a simple substitution cipher (Caesar Cipher).

"Different times, same need — keep messages from the wrong eyes."

2. The Telegraph Era (1830s – 1900)

Faster communication created a greater need for stronger ciphers.



- The electric telegraph revolutionized communication, sending messages across long distances.
- Armies and governments needed ciphers to protect military and diplomatic messages.
- Simple codes gave way to more complex ciphers, such as the Playfair Cipher (1854).

"As messages travelled faster, so did the need for secrecy."

3. Classical Cryptography (1900 – World War I)

More sophisticated methods for a more connected world.



- Ciphers like Playfair, Vigenère, and Vernam were widely used.
- Diplomats and military forces relied on manual encryption and codebooks.
- Cryptanalysis (code-breaking) also advanced, leading to an ongoing "arms race" between code-makers and code-breakers.

"Better ciphers led to stronger challenges — and sharper minds to break them."

4. World Wars & Mechanical Machines (1914 – 1945)

Machines changed the game.

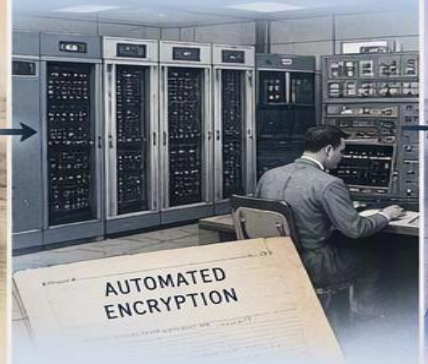


- World War I saw increased use of codebooks and machine-assisted encryption (e.g., the German Enigma began in the 1920s).
- In World War II, mechanical encryption machines like Enigma (Germany), Lorenz (Germany), and Purple (Japan) were widely used.
- Code-breaking efforts, such as Bletchley Park, showed the power of mathematics, logic, and early computers.

"Machines made ciphers stronger — and breaking them became a battle of minds and machines."

5. Early Computing (1945 – 1970s)

From gears to computers.



- The rise of electronic computers enabled new approaches to encryption and cryptanalysis.
- In the 1970s, the Data Encryption Standard (DES) was developed, marking the beginning of modern machine cryptography.

"Computers turned cryptography into a new era of possibility."

6. Modern Machine Cryptography (1980s – Today)

Stronger algorithms for a digital world.



- Public-key cryptography (e.g., RSA, 1977) revolutionized secure communication.
- Advanced algorithms like AES (2001) secure data worldwide.
- Cryptography now protects email, online banking, e-commerce, cryptocurrencies, and almost every digital interaction.

"From telegraph wires to the internet — cryptography keeps the world connected and secure."

SCROLLS → TELEGRAPH → MANUAL CIPHERS → MECHANICAL MACHINES → COMPUTERS → A SECURE DIGITAL TOMORROW

TECHNOLOGY CHANGES. TRUST ENDURES.

Credit: Copilot generated infographic, 2026

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Evolution of Cryptography

The History of Cryptography
From Telegraph to Machine Cryptography
Different tools. Same goal. Private communication.

1830s — 1860s

The Telegraph

The electric telegraph changed communication, making it faster and farther. But it also created a new problem — how to keep messages secret.



Morse code enabled long-distance communication, and the need for coded messages grew quickly.

1860s — 1900s

Code Books & Ciphers

Governments and militaries began using substitution ciphers and code books to protect messages. These methods replaced letters or words with other symbols or codes.



Examples: Simple substitution, keyword ciphers, and one-time pads (early form).

1914 — 1945

World Wars and Advanced Cryptanalysis

World War I and II saw a major arms race in cryptography. Both sides used more complex ciphers, and cryptanalysts worked to break them.



The breaking of the German Enigma code in WWII was a turning point, showing the power of cryptanalysis.

1941 — 1945

The Enigma Machine

The Enigma machine used a complex rotor system to encrypt messages. It was advanced for its time — but not unbreakable.



The Enigma showed both the power and the limits of machine-based cryptography.

1945 — 1970s

Post-War: The Rise of Machine Cryptography

After the war, both the U.S. and the Soviet Union developed machine cryptography systems for secure military and diplomatic communication.



Examples: SIGABA (U.S.), **Fialka** (Soviet Union), and other mechanized systems.

1970s — 1990s

Digital Cryptography and Computers

The advent of computers brought new possibilities. Cryptography evolved from mechanical devices to software and algorithms. Public key cryptography and digital signatures emerged.



Algorithms like RSA (1977) changed how we think about secure communication.

2000s — Today

Modern Cryptography

Today, cryptography is everywhere — in your phone, online banking, Wi-Fi, and the cloud. It protects data at rest, in transit, and in use.



Advanced encryption standards (AES), TLS, and end-to-end encryption keep our digital lives secure.

The Ongoing Journey

From Machines to Quantum and Beyond

The story isn't over. With quantum computing, new challenges and opportunities arise. Cryptography continues to evolve — from classical machines to quantum-resistant algorithms.



The future of cryptography will combine mathematics, technology and innovation to stay ahead of new threats.

From dots and dashes... → *...to algorithms and the cloud.*

Same mission.
Stronger tools.
A more secure tomorrow.

AI-generated infographic to depict the evolution of cryptography

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What is Security?

- Security refers to the measures and protocols put in place to protect data, systems, and networks from unauthorised access, disruption, and damage.

- Key aspects of security include:
  - Protecting sensitive information
  - Ensuring system integrity and availability
  - Preventing unauthorised access.

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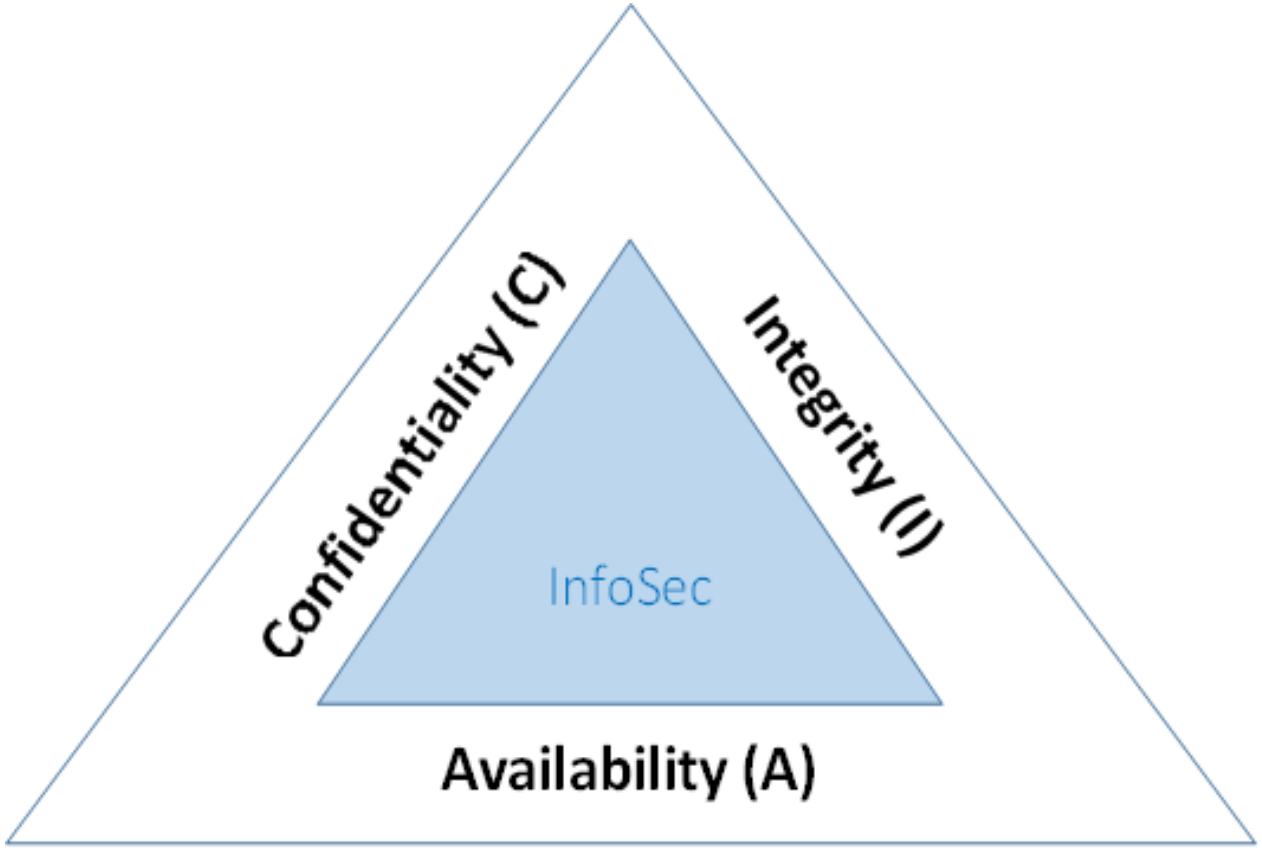
Cybersecurity, Information Security and Network Security

- Cybersecurity is the protection of information that is stored, transmitted, and processed in a networked system of computers, other digital devices, and network devices and transmission lines, including the Internet.
- Cybersecurity includes:
  - Information security
    - The preservation of confidentiality, integrity, and availability of information, to the right people, in the correct form, anywhere, anytime when needed.
  - Network Security
    - The protection of networks and their service from unauthorised modification, destruction, or disclosure, and provision of assurance that the network performs its critical functions correctly and there are no harmful side effects.

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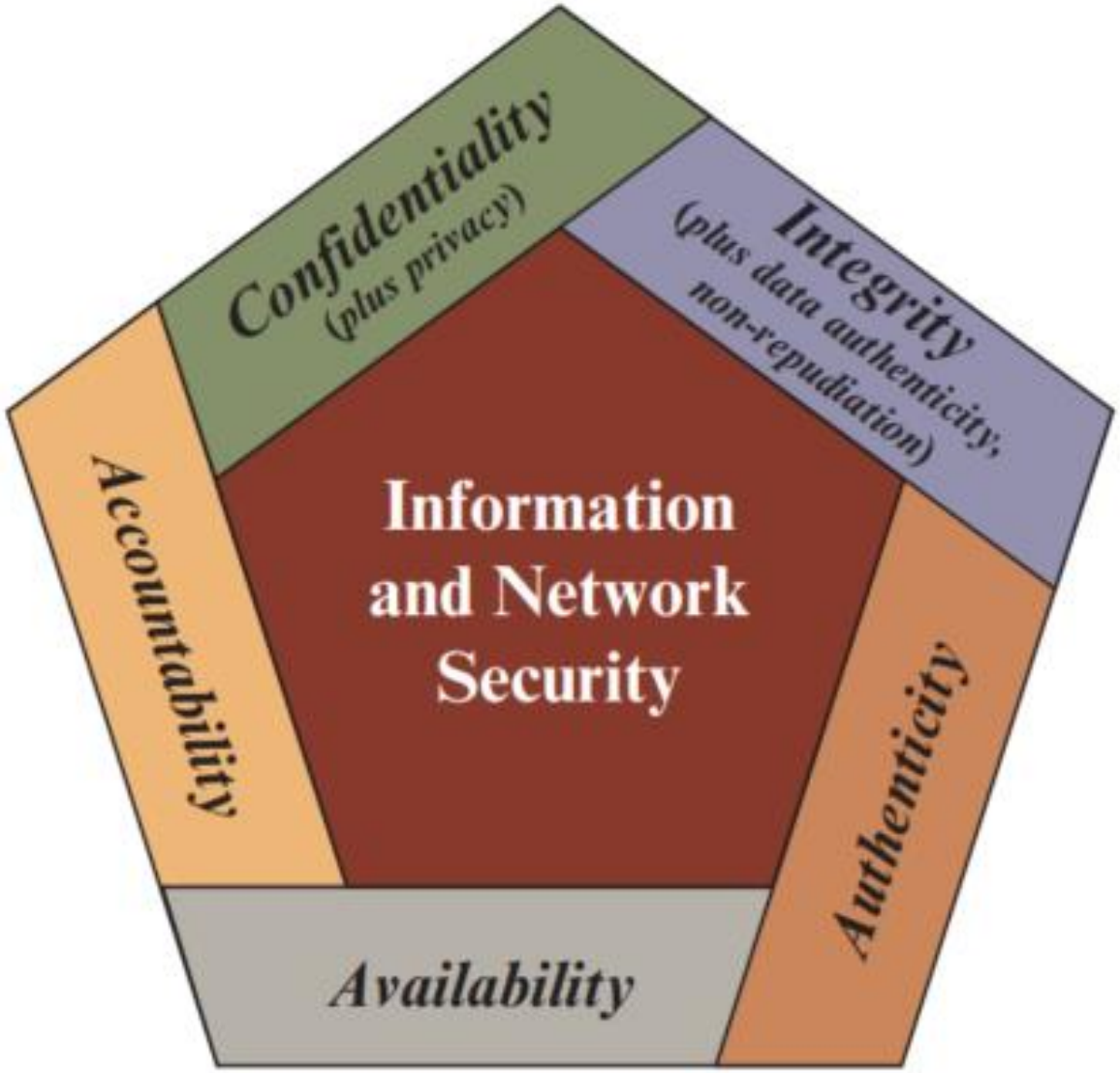
The goal of information security

This is at the heart of information security: Cyber security is a subset of information security.



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Information and Network Security Objectives



Credit: Stallings, 2022

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Cybersecurity, Information Security and Network Security

- Cybersecurity protection properties:
  - Confidentiality
  - Integrity
  - Availability
  - Authenticity
  - Accountability
  - Non-repudiation
- Methods of protection:
  - Organisational policies and procedures,
  - Technical implementation: Encryption
  - Secure communications protocols

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Cybersecurity Protection Properties

- **Confidentiality:** Ensures that sensitive information is only accessible to authorised users. A loss of confidentiality is the unauthorised disclosure of information.
- **Integrity:** Ensures that data remains accurate and unaltered including acts such as improper information modification or destruction. A loss of integrity is the unauthorised modification or destruction of information.
- **Availability:** Ensuring timely and reliable access to and use of information. A loss of availability is the disruption of access to or use of information or an information system.

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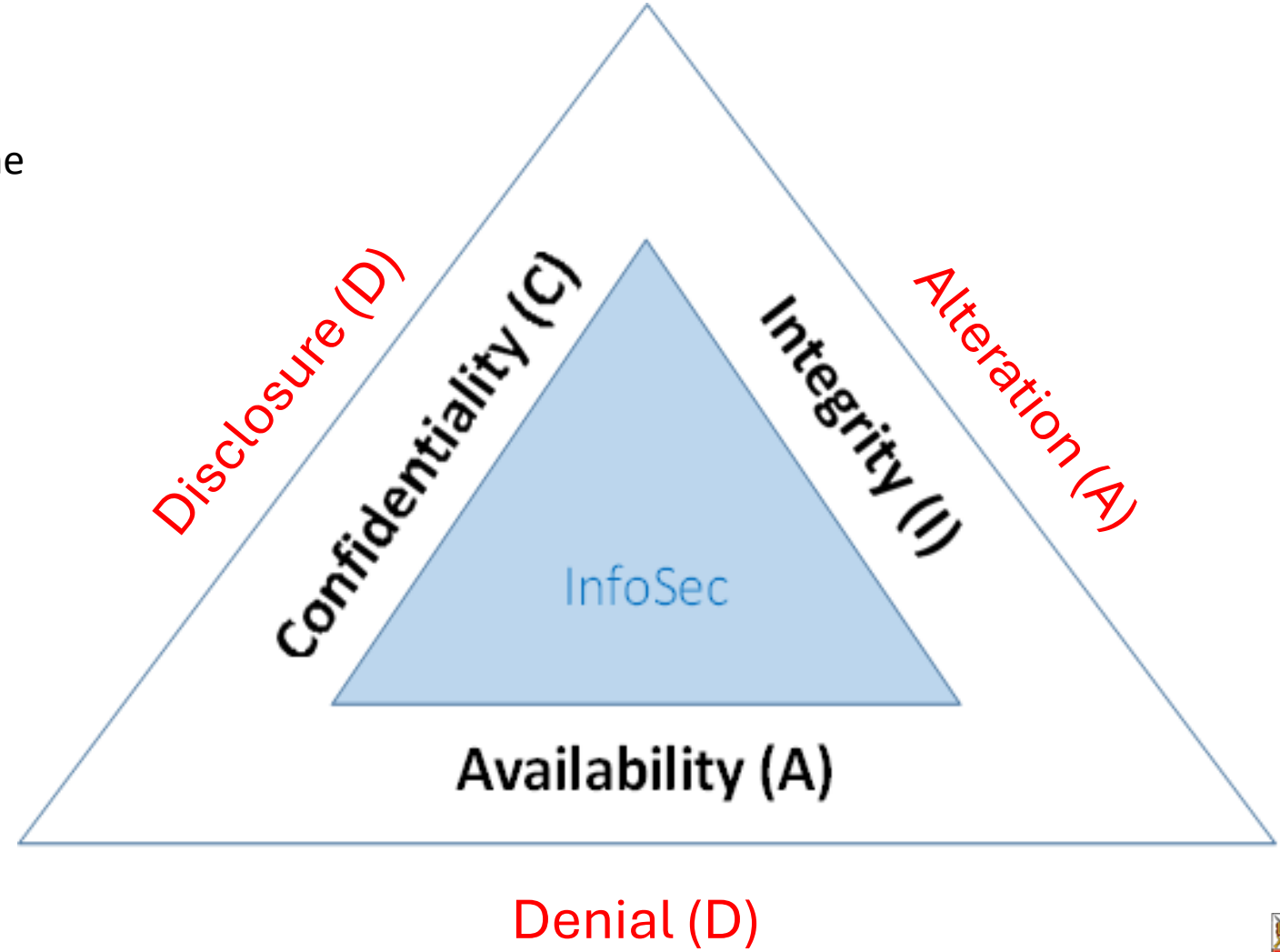
Cybersecurity Protection Properties

- **Authenticity:** The property of being genuine and being able to be verified and trusted; confidence in the validity of a transmission, a message, or message originator.
- **Accountability:** The security goal that generates the requirement for actions of an entity to be traced uniquely to that entity.
- **Non-repudiation:** Assurance that the sender of information is provided with proof of delivery and the recipient is provided with proof of the sender's identity, so neither can later deny having processed the information.
  - Repudiation threats are associated with users who deny performing an action without other parties having any way to prove otherwise.

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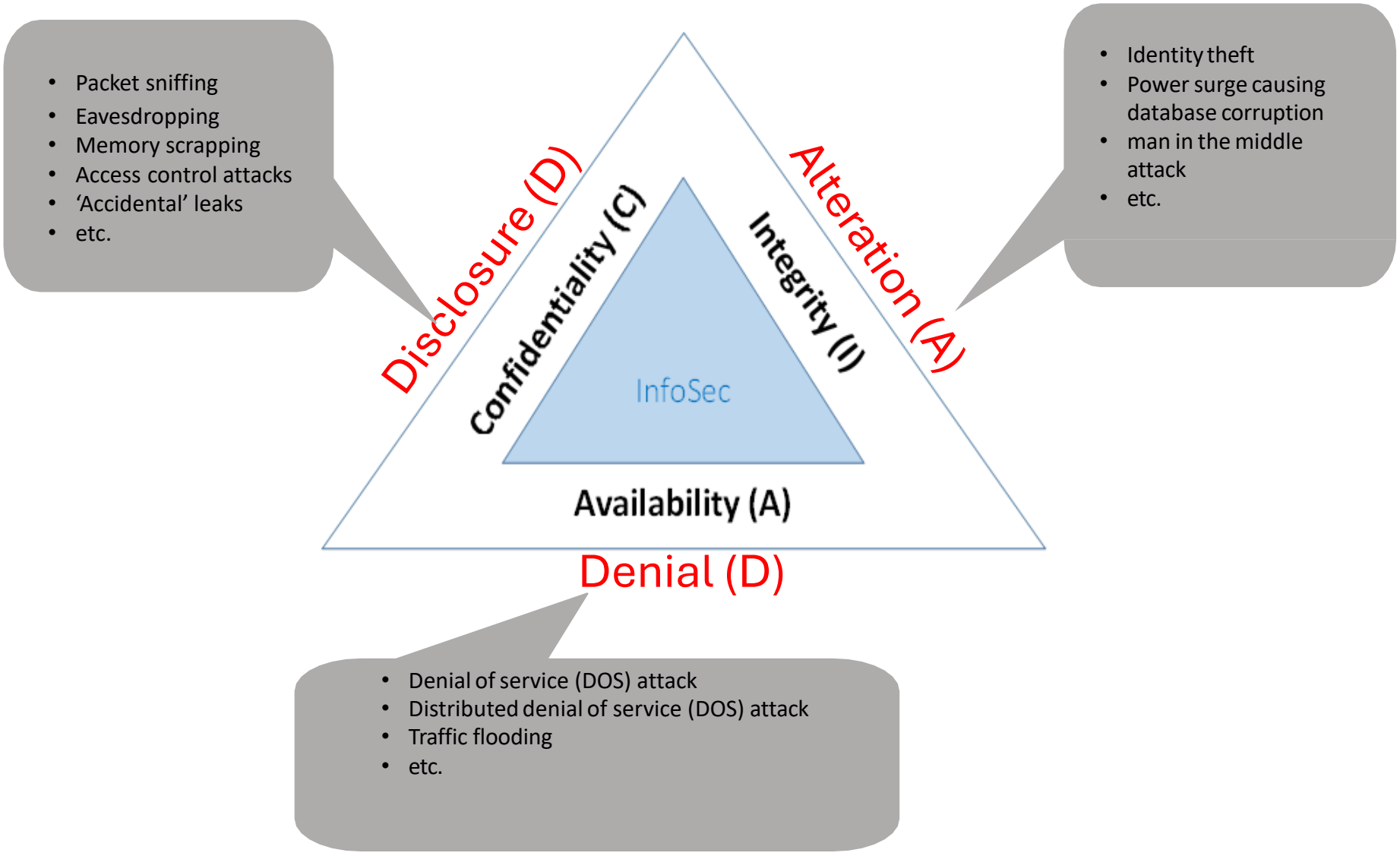
The goal of attackers

Adversaries (attackers / hackers) seek to defeat the CIA. Their goal is DAD.



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The goal of attackers



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DAD Model (Threats to Security)

The DAD model describes threats to security:

- Disclosure: Unauthorised access to confidential information.
- Alteration: Unauthorised modification of information.
- Destruction: Unauthorised deletion or destruction of information.

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Why we need security

- **Protection of Sensitive Data**: Security ensures that confidential and personal information is protected from unauthorised access, misuse, and theft.
- **Preventing Financial Loss**: Data breaches, hacks, and cyberattacks can result in significant financial losses for organisations through stolen information, ransomware, and regulatory penalties.
- **Ensuring Privacy**: Security mechanisms safeguard the privacy of individuals and businesses by preventing unauthorised access to private data.

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Why we need security

- **Maintaining Trust**: Organisations that implement strong security measures build trust with their customers, partners, and employees by ensuring that their data is protected.
- **Compliance with Regulations**: Many industries are required by law to implement specific security standards to protect data (e.g., GDPR, HIPAA). Failure to comply can result in legal consequences.
- **Protecting Intellectual Property**: Security helps organisations safeguard proprietary information and trade secrets from competitors and malicious actors.

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Types of Cyber Crime:

1. **Hacking**: Unauthorised access to data, networks, or systems.
2. **Phishing**: Fraudulent attempts to obtain sensitive information, such as usernames, passwords, or credit card details, by impersonating a trusted entity.
3. **Ransomware**: Malicious software that locks or encrypts data until a ransom is paid.
4. **Identity Theft**: Stealing personal information to commit fraud.
5. **Distributed Denial of Service (DDoS) Attacks**: Flooding a server with traffic to make it unavailable to users.
6. **Cyber Espionage**: Unauthorised access to confidential information for political or commercial gain

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Threat Sources

1. Internal Threats:

- **Disgruntled Employees**: Current or former employees with authorised access who misuse it for malicious purposes.
- **Human Error**: Accidental actions by employees, such as misconfigurations, accidentally deleting data, or falling for phishing attacks.

2. External Threats:

- **Hackers**: Individuals or groups attempting unauthorised access to systems for personal, financial, or political reasons.
- **Cybercriminal Organisations**: Well-organised groups that execute sophisticated cyberattacks for financial gain or to steal sensitive data.
- **Competitors**: Companies may engage in corporate espionage to gain access to proprietary information or trade secrets.

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Threat sources

3. **Nation-State Actors**:
   - **State-Sponsored Attacks**: Cyberattacks launched by government-backed entities targeting other governments, critical infrastructure, or corporations for espionage or disruption.

4. **Natural Threats**:
   - **Natural Disasters**: Earthquakes, floods, fires, or power outages that can disrupt systems and destroy data if not properly backed up.
   - **Pandemics**: Widespread illness affecting personnel availability and operational capabilities, leading to potential vulnerabilities.

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Threat sources

5. Third-Party Providers:

- **Vendors**: Organisations that rely on external vendors or cloud services can be vulnerable to weaknesses in their vendor's security practices.
- **Supply Chain Attacks**: Cyberattacks targeting suppliers and contractors, leveraging their access to larger systems.

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OSI Security Architecture: Attacks, Mechanisms, & Service

- **Security attack:** Any action that compromises the security of information owned by an organisation.
- **Security mechanism:** A process (or a device incorporating such a process) that is designed to detect, prevent, or recover from a security attack.
- **Security service:** A processing or communication service that enhances the security of the data processing systems and the information transfers of an organisation.
- **Threat:** Any circumstance or event with the potential to adversely impact organisational operations (including mission, functions, image, or reputation), organisational assets, individuals, other organisations, or the Nation through an information system via unauthorised access, destruction, disclosure, modification of information, and/or denial of service.
- **Attack:** Any kind of malicious activity that attempts to collect, disrupt, deny, degrade, or destroy information system resources or the information itself.

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OSI Security Architecture: Attacks

Types of Security Attack:

- Passive Attacks:

A passive attack attempts to learn or make use of information from the system but does not affect system resources.

Example: Eavesdropping on, or monitoring of, transmissions.

- Active Attacks: An active attack attempts to alter system resources or affect their operation.

Category:

- Replay
- Masquerade
- Modification of messages
- Denial of service.

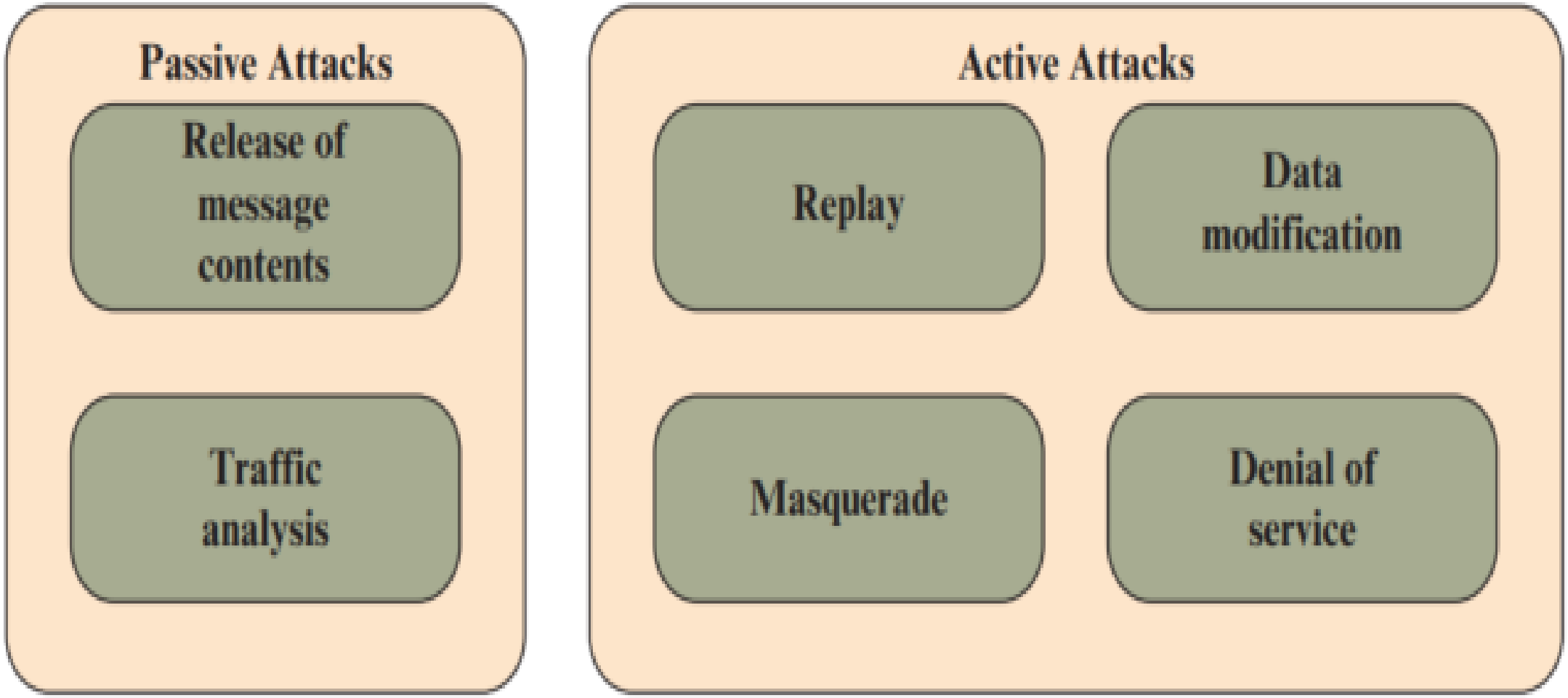
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Active Attacks

- **Masquerade** takes place when one entity pretends to be a different entity. A masquerade attack usually includes one of the other forms of active attack.
- **Replay** involves the passive capture of a data unit and its subsequent retransmission to produce an unauthorised effect.
- **Data modification** occurs when a legitimate message being altered, or that messages are delayed or reordered, to produce an unauthorised effect.
- **Denial of service** prevents or inhibits the normal use or management of communication facilities.

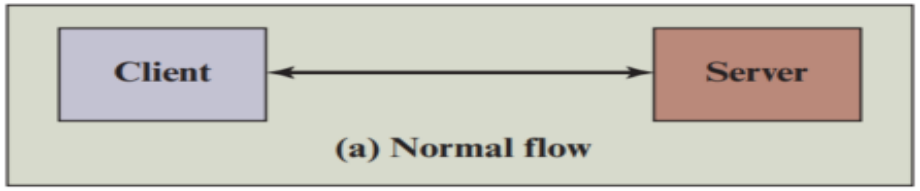
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OSI Security Architecture: Attacks

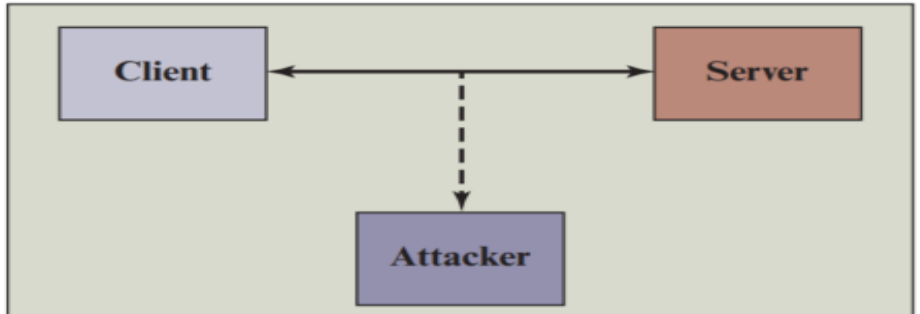


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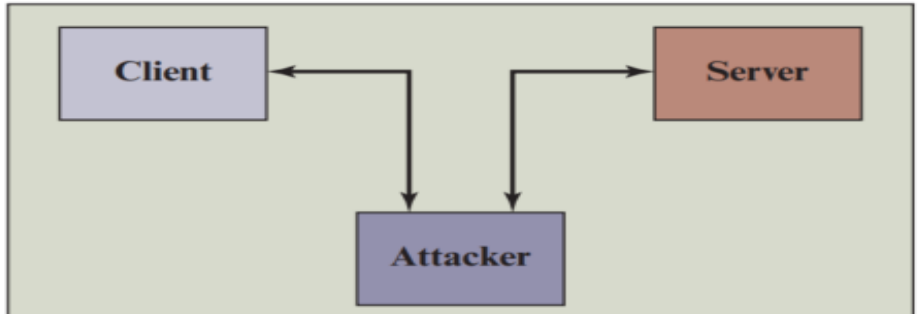
Security Attacks



(a) Normal flow



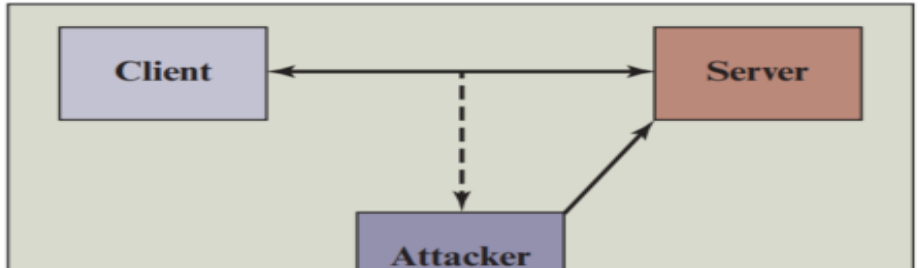
(b) Passive attack (eavesdrop, traffic analysis)



(c) Active attack (masquerade, data modification)



(d) Active attack (masquerade, data modification, denial of service)



(e) Active attack (replay)

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OSI Security Architecture: Mechanism

Security mechanism: A process (or a device incorporating such a process) that is designed to detect, prevent, or recover from a security attack:

Types of security mechanism:

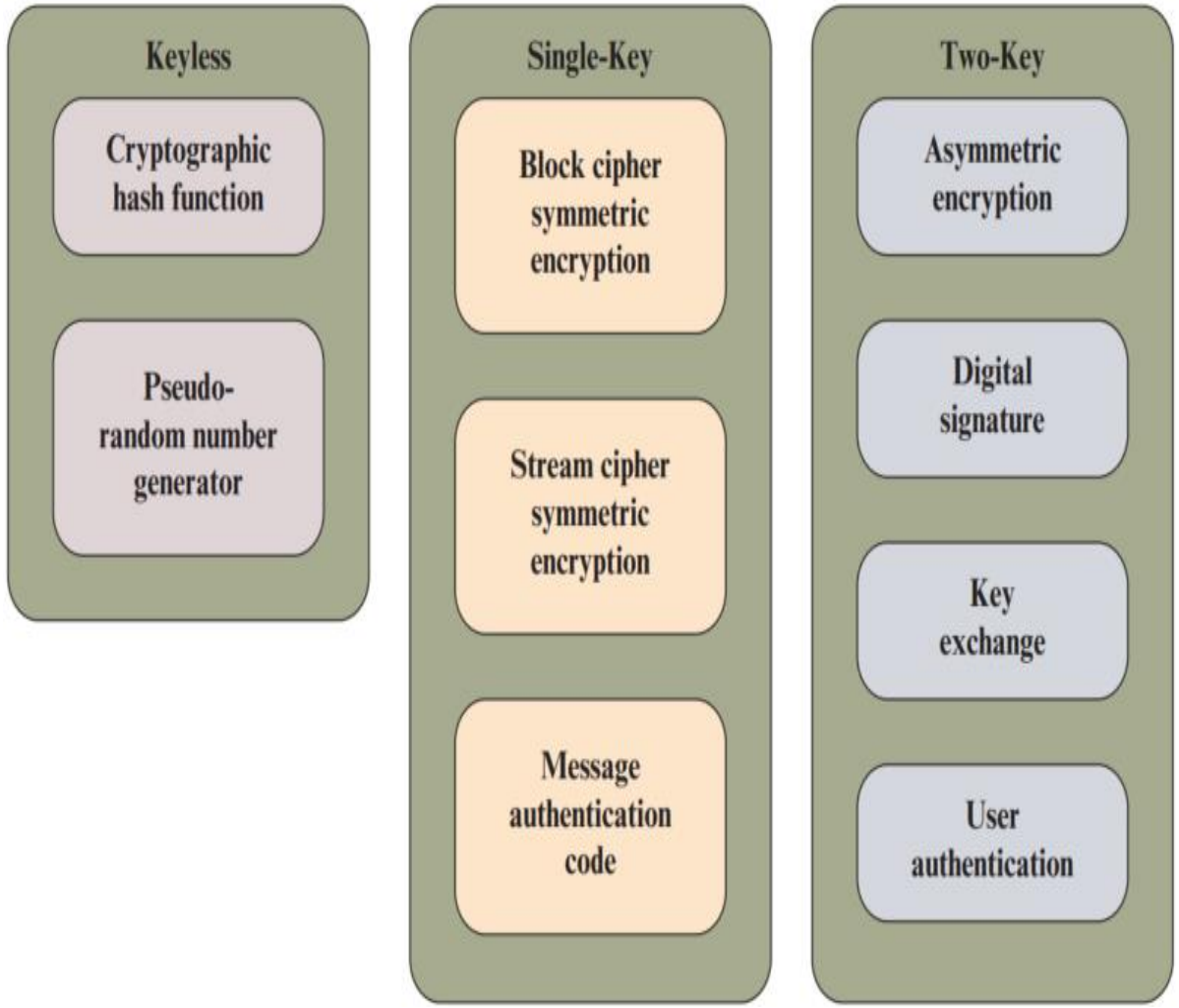
- Cryptographic algorithm
- Data integrity
- Digital signature
- Authentication exchange
- Traffic padding
- Routing control
- Notarisation
- Access control

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Security Services – Cryptographic Algorithm

Cryptographic algorithms can be divided into three categories:

- **Keyless:** Do not use any keys during cryptographic transformations.
- **Single-key:** The result of a transformation is a function of the input data and a single key, known as a secret key.
- **Two-key:** At various stages of the calculation, two different but related keys are used, referred to as a private key and a public key.

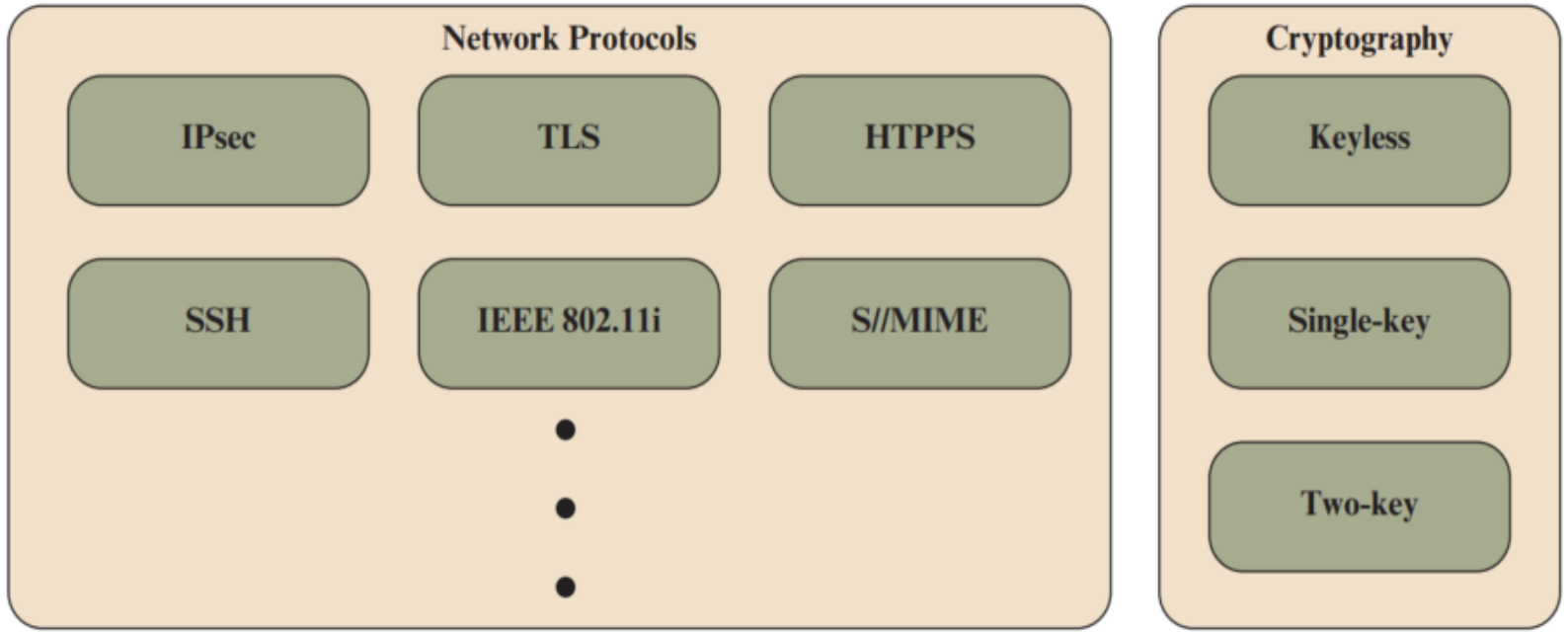


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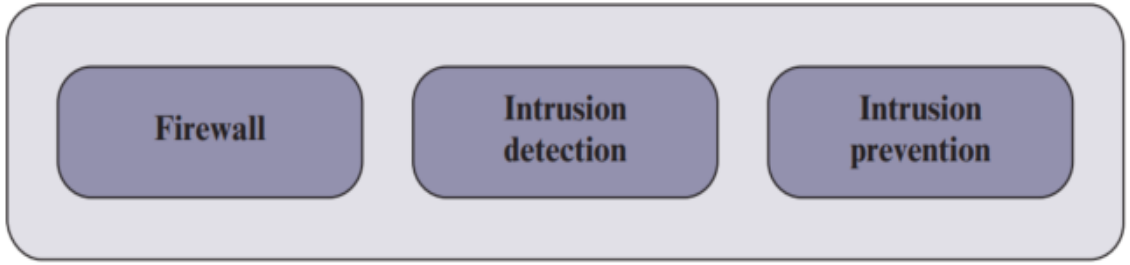
Network Security

Network security consists of:

- Security of the communications pathways of the network
- Security of network devices and devices attached to the network



(a) Communications Security



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Security System

- **Definition**:
  A security system is a framework of policies, tools, and processes designed to protect information, data, networks, and infrastructure from unauthorised access, damage, or attacks.

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Key Components of a Security System:

1. **Access Control**:
   Manages who has access to systems and data based on roles and responsibilities (e.g., multi-factor authentication).
2. **Firewalls**:
   A firewall monitors and controls incoming and outgoing network traffic based on security rules.
3. **Intrusion Detection and Prevention Systems (IDPS)**:
   IDPS monitors network traffic for malicious activity and takes actions to prevent attacks.
4. **Encryption**:
   Converts sensitive data into a coded form to prevent unauthorised access.
5. **Security Policies**:
   Defined rules and guidelines that ensure data protection and regulate user behavior.
6. **Antivirus/Antimalware Software**:
   Detects and removes malicious software that could harm systems or steal data.

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Access control

- A collection of methods and techniques to manage users' access to system resources
  - Supports both the confidentiality and integrity security goals

- Access control follows:
  - Identification
  - Authentication
  - Authorisation

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Access control - Identification

- Identification refers to any unique attribute that can be used to identify a user.
- Examples include:
  - Username
  - Name
  - IP address
  - MAC address

- Note: for the purposes of access management, a user is either a human user or any device that can access other devices/resources.

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Access control - Authentication

- Authentication is any attribute that confirms the identity of the user.
- These are also called the authentication factors and include:

  - Something you know – e.g., password (needs to be strong though), PIN, code, etc.
  - Something you have – etc. card, token, etc.
  - Something you are – e.g., biometric (fingerprint, face, eyes, etc.)

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Strong Passwords

- Do not use things that could be associated with you
- Use Uppercase, Lowercase, Numbers and characters.
- Use every first letter of phrase, then throw in a few numbers
- OmGtii1agtD – how many of you could remember this ?

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Access control – Multifactor Authentication (MFA)

- MFA requires the authentication of users using more than one authentication factor.
- Note that MFA must combine factors from at least two different authentication factors.

- Examples
  - password + token – MFA
  - card + token – not MFA

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Typical Multifactor Authentication (MFA)

- Swiping (or inserting) a card and entering a PIN.
- Logging into a website and being requested to enter an additional one-time password (OTP) that the website's authentication server sends to the requester's phone or email address.
- Downloading a VPN client with a valid digital certificate and logging into the VPN before being granted access to a network.
- Swiping a card, scanning a fingerprint and answering a security question.
- Attaching a USB hardware token to a desktop that generates a one-time passcode and using the one-time passcode to log into a VPN client.

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Contactless Payments

- So why do you think we enabled contactless payments on our debit and credit cards?

- Convenience.
- Often the more secure something is, the less convenient it is to access.

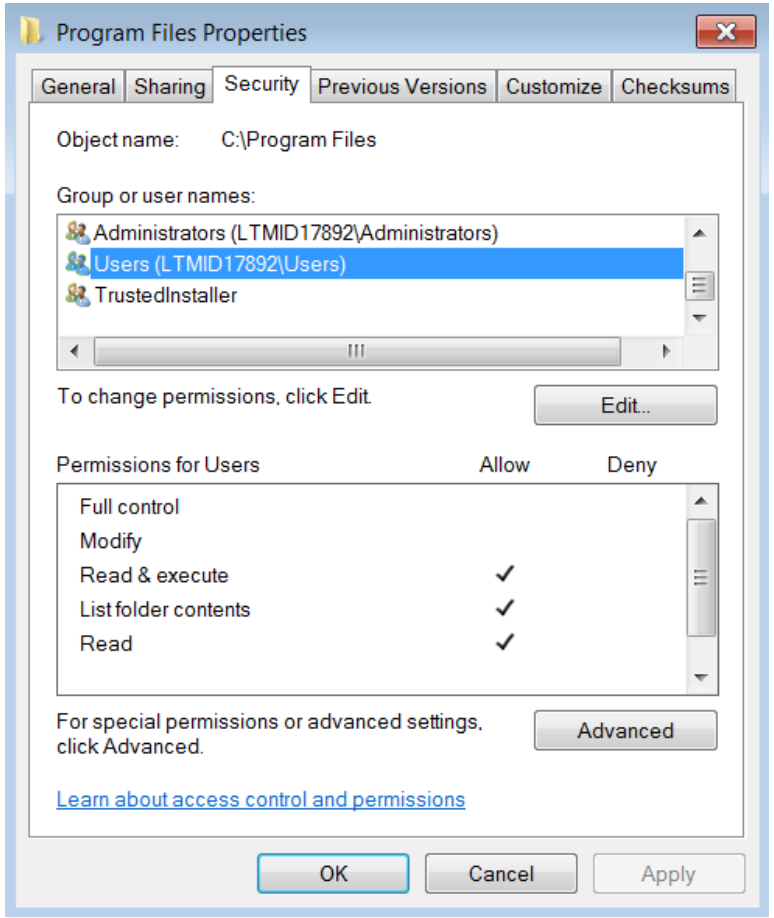
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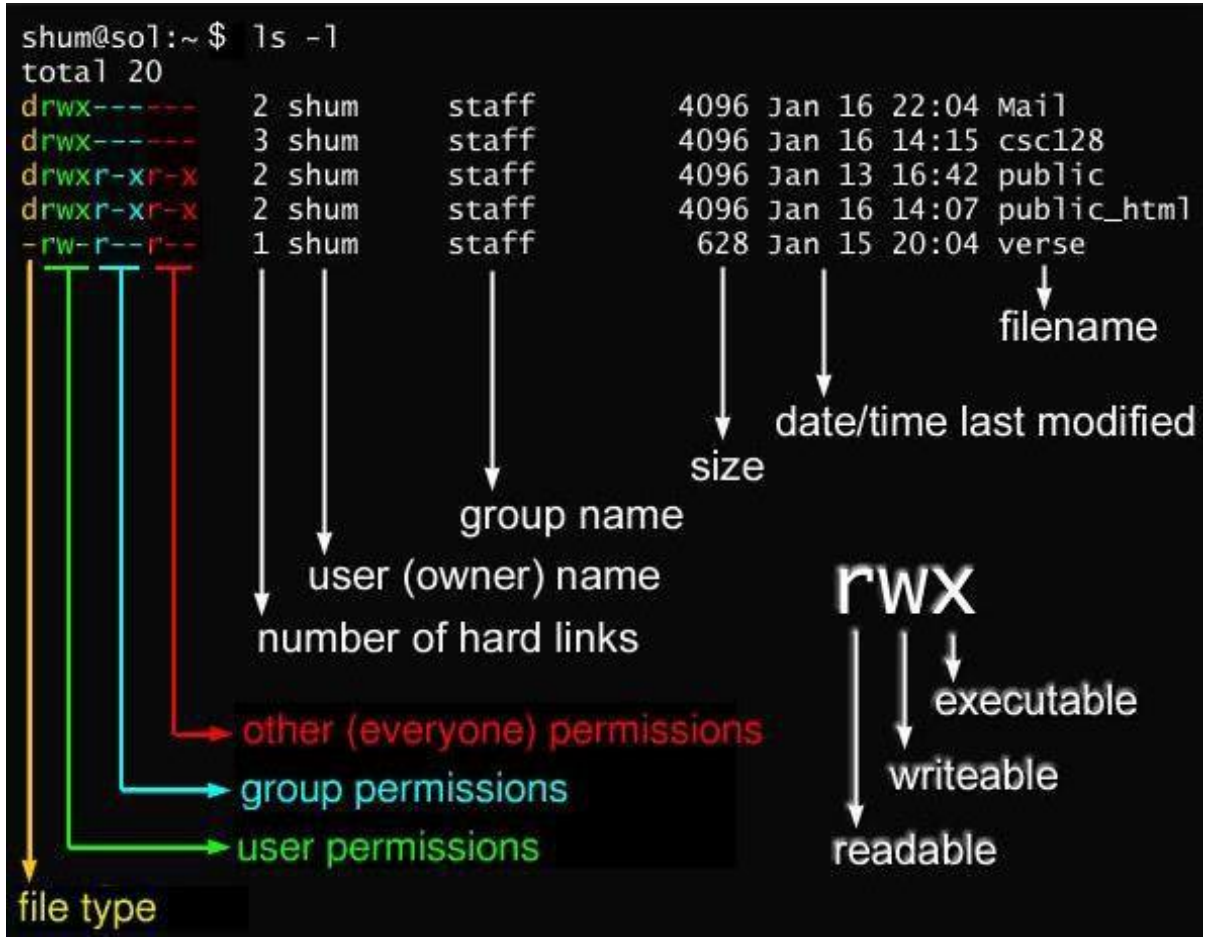
Access Control - Authorisation

- Authorisation determines the level of privilege the user is granted.
- This is usually determined by who they are, their role, the available classifications etc.
- Examples include:
  - Read
  - Write
  - Execute
  - A combination of the above three – rwx
  - Physical restrictions – areas that a user/visitor is allowed to visit
  - Etc.

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Access control – Windows vs Linux





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Access control – Crucial Considerations

- **Least privilege**

Least privilege is based on limiting access to only resources necessary for user's role.

- **SSO - Single Sign-On**

SSO allows the use of one username/identity & password/token/ticket/cert to be used across multiple servers and services that require user authentication.

- **Privilege creep**

The practice of users moving roles and retaining access from previous roles

Q: What are the advantages and disadvantages of the above?

Access control – ID, Authentication & Authorisation

- Take for instance, a visitor arrives at the security gatehouse of a secure site and the security personnel follows this routine:

  - Checks that the visitor's name is on the visitors list for that day – **identification**
  - If name is on the list, requests for a valid ID card to confirm identity – **authentication**
  - If ID checks, issues visitor's card/badge with appropriate restrictions (e.g., a card that can only access card-operated doors in areas to which the visitor is allowed) – **authorisation**

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Physical Security Controls – ISO 27001

These are measures that protect information, information-processing facilities, and supporting infrastructure from unauthorised access, damage, theft, and environmental threats.

- Physical Access Controls Systems
- Perimeter Security
- CCTV and Monitoring Systems

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Physical Access Security Control

These controls help ensure only authorised personnel can enter the secure area.

These controls include:

- Electronic access control (proximity cards, smart cards, biometrics)
- Secure door locks and key management systems
- Turnstiles and mantraps for high-security areas
- Visitor registration and sign-in systems
- Security guards and reception controls

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Physical Access Security Control

1 ELECTRONIC ACCESS CONTROL

Modern credentials for a smarter, safer workplace.



- Proximity cards (RFID)
- Smart cards (encrypted)
- Biometrics (fingerprint, facial recognition, iris scan)
- Integration with access management software
- Audit trails and real-time monitoring

2 SECURE DOOR LOCKS & KEY MANAGEMENT

Reliable hardware. Controlled access. Full accountability.



- Electronic and mechanical door locks
- Networked and wireless lock options
- Key management systems (secure storage, tracking, issuance)
- Restricted key access and audit logs
- Supports multiple door types and facility needs

3 TURNSTILES & MANTRAPS

Controlled entry for high-security areas.



- Tripod and full-height turnstiles
- Mantraps (single or dual interlock)
- Tailgating and piggybacking prevention
- Integration with access control and biometrics
- Ideal for data centers, research facilities, and other high-security areas

4 VISITOR REGISTRATION & SIGN-IN SYSTEMS

Know who's on site. Keep your facility secure.



- Digital visitor registration
- ID verification and badge printing
- Pre-registration and QR code check-in
- Host notification and approval workflows
- Visitor logs and compliance reporting
- Integration with access control and video systems

5 SECURITY GUARDS & RECEPTION CONTROLS

A visible presence. A safer environment.



- On-site security personnel
- Reception desk access control
- Identity verification and credential checks
- Monitoring of visitor and staff movement
- Incident response and emergency support
- Professional and customer-friendly presence

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Perimeter Security

These controls are required to protect physical security perimeters around sensitive areas.

Types:

- Fencing and security barriers
- Security gates
- Bollards and vehicle-access controls
- Security lighting around buildings
- Secure windows and reinforced door

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Perimeter Security

1 FENCING AND SECURITY BARRIERS
Defend the boundary. Deter unauthorized entry.



- Perimeter fencing (chain link, weld mesh, palisade)
- Anti-climb and anti-cut designs
- Razor wire and electric fencing options
- Integration with intrusion detection sensors



4 SECURITY LIGHTING AROUND BUILDINGS
Illuminate. Deter. Improve visibility.



- High-intensity LED lighting
- Motion-activated sensors
- Covers perimeter, entry points and dark areas
- Integrates with security systems (CCTV, alarms)

2 SENCURITY GATES
Control access. Keep unauthorized vehicles out.



- Sliding or swing gates (manual or automatic)
- Access control integration (cards, RFID, ANPR)
- Vehicle and pedestrian gates
- Intercom and video verification
- Failsafe and emergency release options

3 BOLLARDS AND VEHICLE-ACCESS CONTROLS
Stop threats before they reach your facility.



- Fixed, removable or automatic bollards
- Crash-rated options for high-security sites
- Vehicle barriers (boom gates, road blockers, tire spikes)
- Integrated with access control and traffic management
- Prevents unauthorized or ramming vehicle entry

5 SECURE WINDOWS AND REINFORCED DOORS
Strengthen entry points. Resist forced access.



- Impact-resistant and laminated security windows
- Reinforced doors with high-security locks
- Blast, bullet or forced-entry resistant options (as required)
- Tamper-resistant frames and hardware
- Ideal for critical areas (control rooms, data centers, admin offices)

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CCTV and Monitoring Systems

Physical security monitoring is specifically recommended to detect unauthorised access or suspicious activity.

Categories:

- CCTV surveillance with recording capability
- Intruder alarm systems
- Motion detectors
- Security monitoring and incident alerting systems

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CCTV and Monitoring Systems

1 CCTV SURVEILLANCE WITH RECORDING CAPABILITY

Continuous video monitoring and recording for evidence, investigation and deterrence.



- High-definition video capture
- 24/7 recording and storage
- Indoor and outdoor camera options
- Remote viewing (desktop/mobile)
- Supports investigation and compliance

2 INTRUDER ALARM SYSTEMS

Detect and alert on unauthorized access to protect people and assets.



- Monitors doors, windows and sensitive areas
- Instant alerts on unauthorized entry
- Audible and visual alarms
- Integration with CCTV and monitoring systems
- Remote arming/disarming

3 MOTION DETECTORS

Sense movement and trigger alerts when activity is detected.





Ceiling-mounted



Wall-mounted



Outdoor (weatherproof)

- Detects movement in protected areas
- Indoor and outdoor options
- Reduces false alarms with smart detection
- Triggers alarms, lighting or CCTV recording
- Adjustable sensitivity and coverage

4 SECURITY MONITORING AND INCIDENT ALERTING SYSTEMS

Centralized monitoring, real-time alerts and faster response to security incidents.

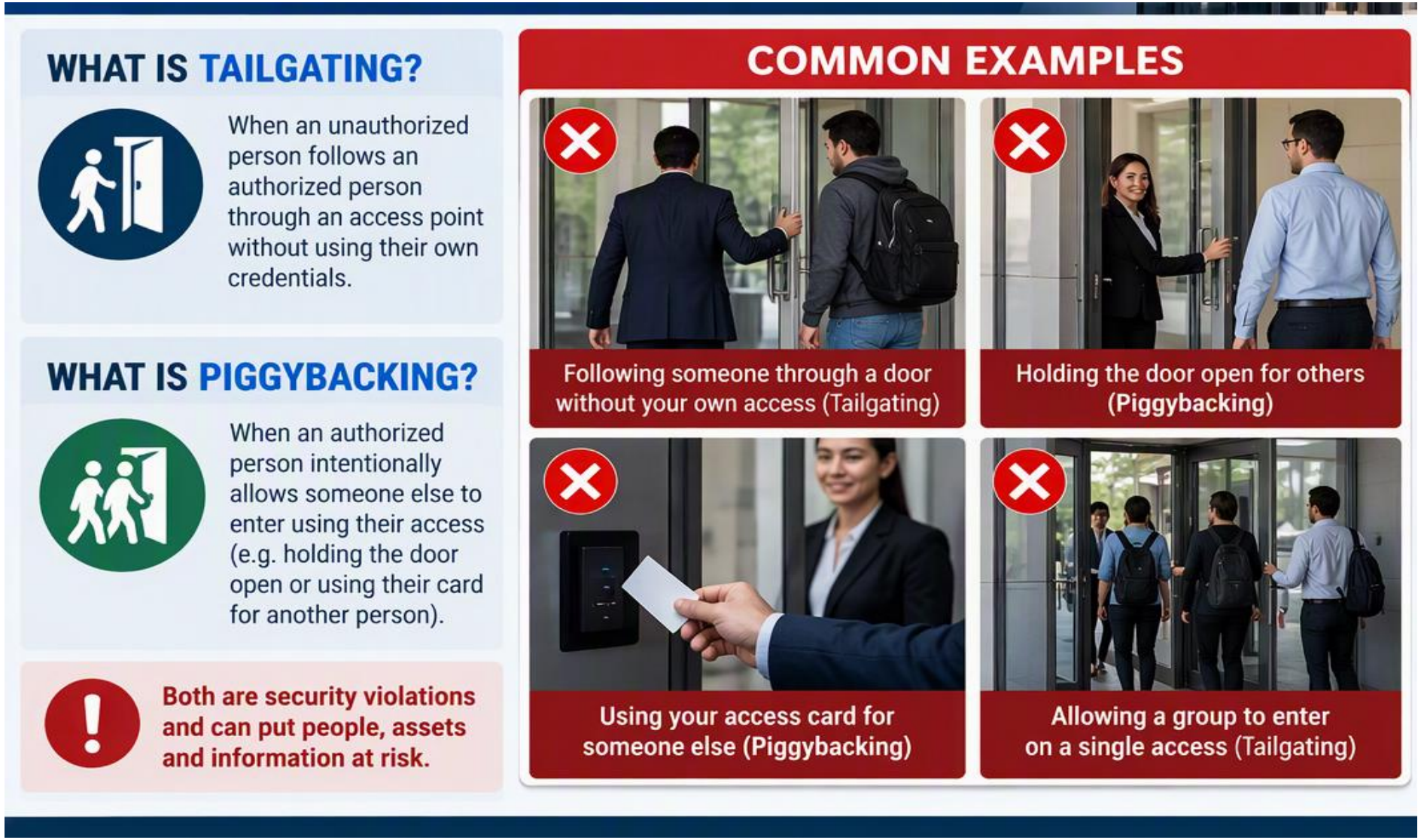


- 24/7 monitoring (on-site or remote)
- Real-time incident alerts (SMS, email, app)
- Integration with CCTV, alarms and sensors
- Event logs and alarm management
- Supports faster response and escalation

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Access control – Tailgating and Piggybacking



Both are similar but tailgating is without consent of an authorised person while piggybacking is with consent of an authorised person.

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Physical Security

- What about physical security vulnerabilities?
- Often less considered
- Discuss the Human Factor
- What are the risks?

Human Factors - Not always malicious

1. Distraction
   - Distraction leads people to miss goals due to a lack of focus. Not paying attention or not allocating sufficient resources results in frustration and time wasted. Distracted people tend not to spend the right amount of energy and concentration while performing relevant tasks
2. Lack of Control
   - Typical of impulsive people. Not having self-control means being a victim of events, acting irrationally as a result.
3. Ignorance and Overconfidence
   - Ignorance and overconfidence lead people to take risks. This, while they do not have the knowledge or skills required to manage and deal with the likely disastrous consequences.

Questions

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